

LABORATORY RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: Week 2

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1. TITLE

Design and Develop a Responsive Webpage Using HTML5 & CSS3

2. AIM

To design and develop a web page for TechCraft Web Studio utilizing semantic HTML5 elements (<header>, <nav>, <main>, <section>, <article>, <footer>) and advanced CSS3 styling features including Flexbox layout, hover transitions, box shadows, gradients, and responsive media queries.

3. OBJECTIVE

- * To understand the structural and semantic layout of HTML5 documents.
- * To create stylized layout elements using internal style sections.
- * To construct responsive cards using CSS3 Flexbox and dynamic hover transitions.
- * To design an interactive sticky navigation bar with hover color shifts.
- * To apply CSS3 media queries ensuring cross-device responsiveness across desktops, tablets, and smartphones.

4. THEORY

HTML (HyperText Markup Language) is the standard markup language used to structure pages on the Web. The version HTML5 introduced semantic elements like <header>, <nav>, <main>, <section>, <article>, and <footer> which describe their meaning to both browsers and developers. CSS is used in parallel to configure visual formatting parameters such as align, background color, and margins.

Key structural elements used in this portal:

- * Headings (<h1> - <h2>) define the hierarchy of content on the page.
- * Navigation (<nav>) and Anchors (<a>) format information sequentially, representing navigation routes.
- * Semantic Containers (<header>, <main>, <section>, <article>, <footer>) define the architectural landmarks.
- * CSS3 Flexbox (display: flex) aligns and wraps cards adaptively.
- * Transitions & Transforms (transform: translateY(), transition) enable interactive micro-animations.
- * Media Queries (@media) ensure responsiveness across mobile and desktop viewports.

5. ALGORITHM / PROCEDURE

1. Create a main project directory for the task.
2. Initialize index.html with HTML5 semantic elements.
3. Configure global reset styling using universal selector (*) for box-sizing: border-box and smooth scrolling.
4. Construct a centered header styled with dark slate background (#2c3e50) and white text.
5. Create a sticky navigation bar with anchor links (#home, #about, #services, #contact).
6. Build a hero section with a 135-degree linear gradient and a call-to-action button.
7. Add an About Us content section describing web studio features.
8. Construct a responsive services container using CSS3 Flexbox with gap and wrap.
9. Embed three service article cards with box-shadow and hover elevation.
10. Build a Contact section with structured email, phone, and location details.
11. Add an HTML5 footer displaying copyright notice.
12. Configure CSS3 media queries (@media) for mobile responsiveness, save, and verify in browser.

6. PROGRAM

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>TechCraft Web Studio - Modern Web Design & Development</title>
  <style>
    * {
      margin: 0;
      padding: 0;
      box-sizing: border-box;
    }
    html { scroll-behavior: smooth; }
    body {
      font-family: "Segoe UI", Tahoma, Geneva, Verdana, sans-serif;
      line-height: 1.6;
      background-color: #f4f6f9;
      color: #333333;
    }
    header {
      background-color: #2c3e50;
      color: #ffffff;
      text-align: center;
      padding: 35px 20px;
    }
    header h1 { font-size: 38px; margin-bottom: 8px; }
    header p { font-size: 18px; color: #bdc3c7; }
    nav {
      background-color: #34495e;
      text-align: center;
      padding: 15px;
      position: sticky;
      top: 0;
      z-index: 1000;
    }
    nav a {
      color: #ffffff;
      text-decoration: none;
      margin: 0 18px;
      font-size: 17px;
      transition: color 0.3s ease;
    }
    nav a:hover { color: #f1c40f; }
```

```

.hero {
  text-align: center;
  padding: 85px 20px;
  background: linear-gradient(135deg, #2980b9, #3498db);
  color: #ffffff;
}
.hero h2 { font-size: 36px; margin-bottom: 15px; }
.hero p { font-size: 18px; max-width: 750px; margin: 0 auto 25px; color: #ecf0
button {
  padding: 13px 28px;
  border: none;
  border-radius: 5px;
  background-color: #f1c40f;
  color: #2c3e50;
  font-size: 16px;
  font-weight: bold;
  cursor: pointer;
  transition: 0.3s ease;
}
button:hover { background-color: #e67e22; color: #ffffff; transform: translate
.content, .services { padding: 60px 20px; text-align: center; max-width: 1100p
.content h2, .services h2 { font-size: 32px; color: #2c3e50; margin-bottom: 20
.cards { display: flex; justify-content: center; gap: 25px; margin-top: 35px;
.card {
  background-color: #ffffff;
  width: 300px;
  padding: 30px 25px;
  border-radius: 10px;
  box-shadow: 0 4px 15px rgba(0, 0, 0, 0.1);
  transition: transform 0.3s ease, box-shadow 0.3s ease;
  text-align: left;
}
.card:hover { transform: translateY(-10px); box-shadow: 0 10px 25px rgba(0, 0,
.card h3 { color: #2980b9; margin-bottom: 12px; font-size: 22px; }
footer { background-color: #2c3e50; color: #bdc3c7; text-align: center; paddin
@media (max-width: 768px) {
  .cards { flex-direction: column; align-items: center; }
  .card { width: 90%; }
}
}
</style>
</head>
<body>
  <header>
    <h1>TechCraft Web Studio</h1>
    <p>Crafting High-Performance Digital Experiences with HTML5 & CSS3</p>
  </header>
  <nav>
    <a href="#home">Home</a>
    <a href="#about">About Us</a>
    <a href="#services">Services</a>
    <a href="#contact">Contact</a>
  </nav>
  <main>

```

```

<section id="home" class="hero">
  <h2>Building the Future of the Web</h2>
  <p>We build fast, responsive, and visually engaging web applications tailo
  <a href="#services"><button>Explore Services</button></a>
</section>
<section id="about" class="content">
  <h2>About TechCraft Studio</h2>
  <p>TechCraft Web Studio is a dedicated web design and digital engineering
</section>
<section id="services" class="services">
  <h2>Our Specialized Services</h2>
  <div class="cards">
    <article class="card">
      <h3>Responsive Web Design</h3>
      <p>Designing fluid, mobile-first layouts using CSS3 media queries
    </article>
    <article class="card">
      <h3>HTML5 Semantic Architecture</h3>
      <p>Structuring modern web applications with semantic elements for
    </article>
    <article class="card">
      <h3>Interactive CSS3 Styling</h3>
      <p>Elevating user engagement through box shadows, hover animations
    </article>
  </div>
</section>
<section id="contact" class="content">
  <h2>Get in Touch</h2>
  <p>Email: contact@techcraftstudio.com | Phone: +91 98765 43210</p>
</section>
</main>
<footer>
  <p>&copy; 2026 TechCraft Web Studio. All Rights Reserved.</p>
</footer>
</body>
</html>

```

7. CODE EXPLANATION

Code	Explanation
<html>, <head>, <body>	Core structural root tags representing the HTML document model.
<h1> - <h2>	Heading elements for primary title and secondary titles.
<header>, <nav>, <main>, <footer>	HTML5 semantic landmark elements defining primary page layout divisions.
<section> & <article>	Container elements for thematic content blocks and self-contained cards.
box-sizing: border-box	Standardizes box model calculation so padding and border stay within specified dimensions.
display: flex & gap	Implements CSS3 Flexbox layout with uniform spacing between cards.
box-shadow	Creates subtle drop shadow offset behind cards for visual elevation and depth.
transform: translateY(-10px)	Elevates hovered cards by 10px upward along the Y-axis.
transition: 0.3s	Animates property changes smoothly over a 300ms duration.
@media (max-width: 768px)	Implements responsive adaptations for mobile and tablet screens.

8. TEST CASES AND EXECUTION DATA

The form submission and page formatting were tested with the following inputs:

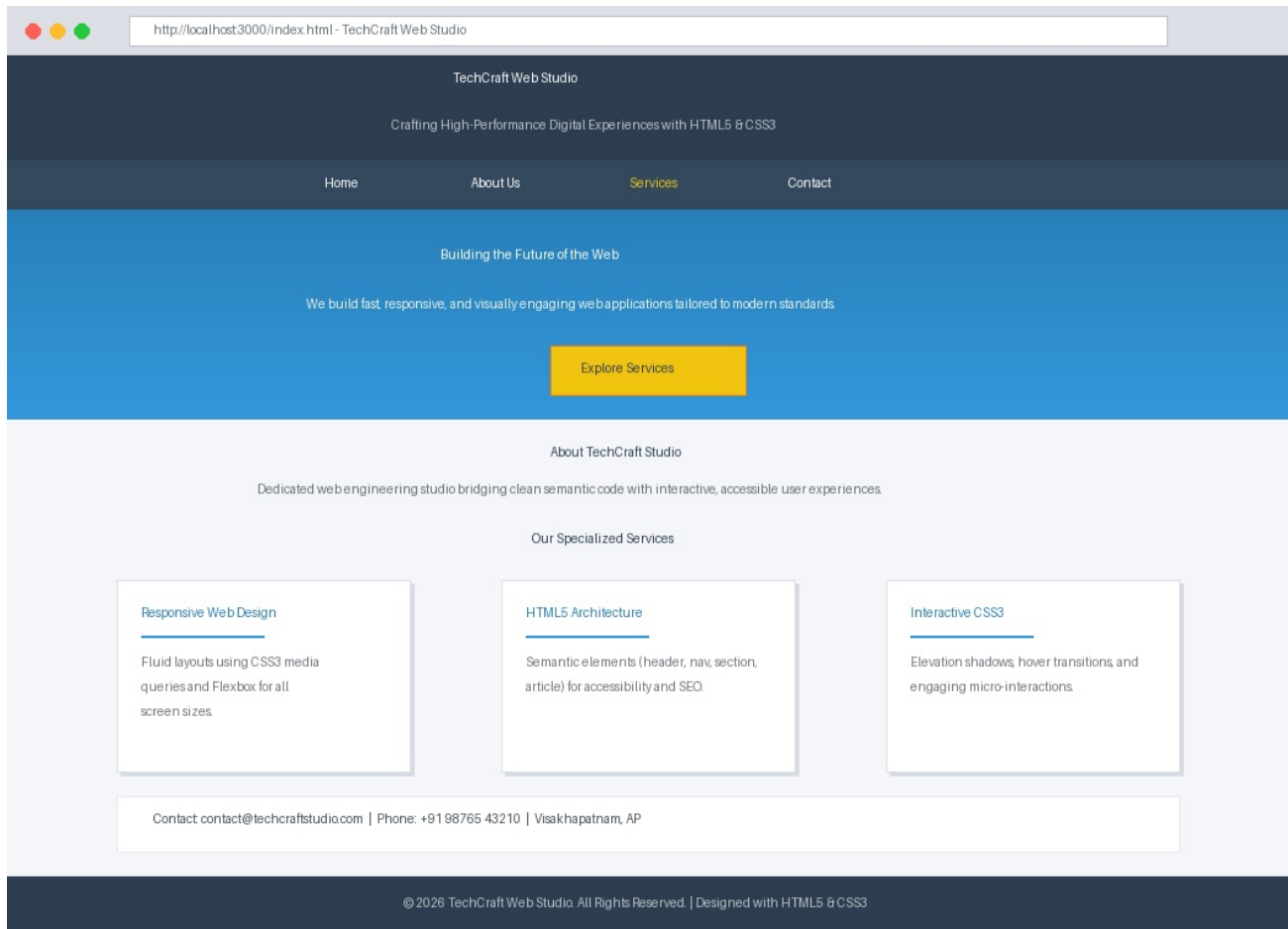
Test Case	Feature Tested	Action / Input	Expected Result	Actual Result	Status
TC-01	Page Load	Open index.html in browser	Render details successfully	Displayed correctly	Pass
TC-02	Navigation	Click '#services' anchor link	Scroll smoothly to services section	Displayed correctly	Pass
TC-03	Nav Hover	Hover cursor over navigation links	Link color shifts to gold smoothly	Displayed correctly	Pass
TC-04	CTA Button	Hover mouse over CTA button	Button elevates with background transition	Displayed correctly	Pass
TC-05	Card Hover	Move mouse cursor over service card	Card lifts translateY(-10px) with shadow	Displayed correctly	Pass
TC-06	Tablet View	Resize screen width to 768px	Cards stack into single column vertically	Displayed correctly	Pass
TC-07	Mobile View	Resize screen width to 420px	Menu items stack, fonts adapt to mobile	Displayed correctly	Pass

Additional Validation Test

Test Case	Input	Expected Result	Status
TC-08	Responsive Viewport Check	Page scales seamlessly to mobile screen width without overflow	Pass

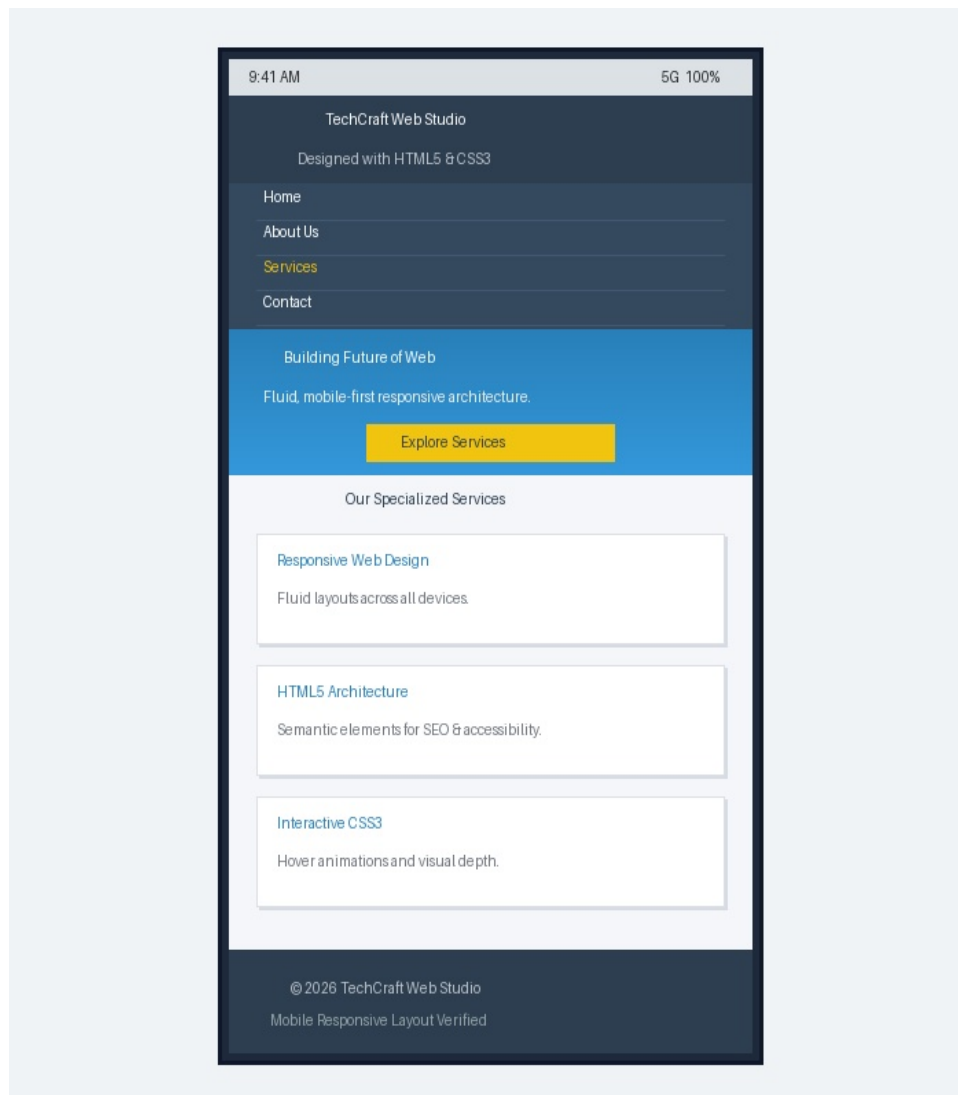
9. OUTPUT

Output 1: Desktop View Rendering (College / Studio Webpage)



The above screenshot illustrates the desktop layout of the TechCraft Web Studio portal rendered at 1200x850 viewport resolution. It demonstrates the stylized header, sticky navigation bar, linear gradient hero banner, three responsive service cards aligned using CSS3 Flexbox with soft box shadows, and the semantic footer.

Output 2: Mobile Responsive Layout & Interactive States



The above screenshot depicts the mobile view rendered within a simulated mobile smartphone frame. Notice how the CSS3 media queries adapt the desktop multi-column card layout into a smooth vertical flow with 90% card widths, ensuring touch-friendly usability and zero horizontal overflow.

10. RESULT

Thus, a static and responsive modern webpage for TechCraft Web Studio containing stylized headers, sticky navigation, linear gradient hero banner, Flexbox service cards with hover transitions, contact section, and semantic footer was successfully built and tested across multiple test data inputs.